

generateStats

Takes a client id as input, and sums up the reservation count, tidal days worth of reservations from that client, and total cost of that clients reservations. It then outputs that to the stats table.

```
CREATE OR REPLACE PROCEDURE GENERATE_STATS (  
  p_C_ID IN NUMBER)  
AS  
  V_TOT_RESERVATIONS INT := 0;  
  V_TOT_DAYS_OF_RESERVATIONS INT := 0;  
  V_TOT_COST_OF_RESERVATIONS INT := 0;  
  V_GENDER CHAR(1);  
  
BEGIN  
  SELECT COUNT(R_ID) INTO V_TOT_RESERVATIONS FROM RESERVATION WHERE C_ID =  
  p_C_ID;  
  DBMS_OUTPUT.PUT_LINE('Total Reservations: ' || V_TOT_RESERVATIONS);  
  
  SELECT SUM(END_DATE - START_DATE) INTO V_TOT_DAYS_OF_RESERVATIONS FROM  
  RESERVATION WHERE C_ID = p_C_ID;  
  DBMS_OUTPUT.PUT_LINE('Total Days Of Reservations: ' ||  
  V_TOT_DAYS_OF_RESERVATIONS);  
  
  SELECT SUM(TOTAL_COST) INTO V_TOT_COST_OF_RESERVATIONS FROM RESERVATION  
  WHERE C_ID = p_C_ID;  
  DBMS_OUTPUT.PUT_LINE('Total Cost of Reservations: ' ||  
  V_TOT_COST_OF_RESERVATIONS);  
  
  SELECT GENDER INTO V_GENDER  
  FROM CLIENT  
  WHERE C_ID = p_C_ID;  
  
  INSERT INTO STATS (YEAR, C_ID, GENDER, TOT_RESERVATIONS,  
  TOT_DAYS_OF_RESERVATIONS, TOT_COST_OF_RESERVATIONS)  
  VALUES (2024, p_C_ID, V_GENDER, V_TOT_RESERVATIONS, V_TOT_DAYS_OF_RESERVATIONS,  
  V_TOT_COST_OF_RESERVATIONS);  
  
END;
```

Procedure GENERATE_STATS compiled

Total Reservations: 2
Total Days Of Reservations: 11
Total Cost of Reservations: 680

findActiveClients

Takes a start and end date range, and returns all of the clients who have\had reservations within that time frame. This can satisfy questions 1, 2, and 4 just by altering the desired date range.

```
CREATE OR REPLACE PROCEDURE FIND_ACTIVE_CLIENTS (
  p_START_DATE IN DATE,
  p_END_DATE IN DATE
) AS
  CURSOR active_clients_cursor IS
    SELECT DISTINCT C.C_ID, C.F_NAME, C.L_NAME, R.START_DATE, R.END_DATE
    FROM CLIENT C
    JOIN RESERVATION R ON C.C_ID = R.C_ID
    WHERE R.START_DATE <= p_END_DATE AND R.END_DATE >= p_START_DATE;

  v_client_id CLIENT.C_ID%TYPE;
  v_first_name CLIENT.F_NAME%TYPE;
  v_last_name CLIENT.L_NAME%TYPE;
  v_start_date DATE;
  v_end_date DATE;
BEGIN
  DBMS_OUTPUT.PUT_LINE('Active Clients for the specified period:');
  DBMS_OUTPUT.PUT_LINE('-----');
  FOR client_record IN active_clients_cursor LOOP
    DBMS_OUTPUT.PUT_LINE('Client ID: ' || client_record.C_ID ||
                          ', Name: ' || client_record.F_NAME || ' ' ||
client_record.L_NAME ||
                          ', Reservation Period: ' ||
client_record.START_DATE || ' to ' || client_record.END_DATE);
  END LOOP;
END;
/
```

Active Clients for the specified period:

Client ID: 2, Name: Jane Smith, Reservation Period: 24-05-01 to 24-05-07

yearlyArchiveReservations

Moves all reservations into the archived reservations table at the end of the year and then removes them from the reservations table.

```
CREATE OR REPLACE PROCEDURE YearlyArchiveReservations AS
BEGIN
    INSERT INTO ARCHIVED_RESERVATION (R_ID, START_DATE, END_DATE, C_ID,
V_ID, TOTAL_COST, ADDED_BY)
    SELECT R_ID, START_DATE, END_DATE, C_ID, V_ID, TOTAL_COST, 'system'
    FROM RESERVATION
    WHERE END_DATE IS NOT NULL
        AND END_DATE < ADD_MONTHS(SYSDATE, -12);

    DELETE FROM RESERVATION
    WHERE END_DATE IS NOT NULL
        AND END_DATE < ADD_MONTHS(SYSDATE, -12);

    COMMIT;

EXCEPTION
    WHEN OTHERS THEN
        ROLLBACK;
        RAISE;
END;
/
```

Archived reservation table after:

	R_ID	START_DATE	END_DATE	C_ID	V_ID	TOTAL_COST	ADDED_BY	ADDED_DATE
1	101	23-01-01	23-01-05	2	1000	200	Admin	23-01-06
2	102	23-02-10	23-02-12	3	1001	300	Admin	23-02-13

blacklistClient

Takes a client id as input and adds them into the blacklisted clients table. Does Not remove them from the clients table.

```
CREATE OR REPLACE PROCEDURE BLACKLIST_CLIENT (  
    p_C_ID IN NUMBER  
)  
AS  
BEGIN  
    INSERT INTO BLACK_LIST (C_ID, F_NAME, L_NAME, DOB, CITY, GENDER,  
    REASON, ADDED_BY, ADDED_DATE)  
    SELECT CLIENT.C_ID, CLIENT.F_NAME, CLIENT.L_NAME, CLIENT.DOB,  
    CLIENT.CITY, CLIENT.GENDER,  
        'Excessive cancellations', 'Admin', SYSDATE  
    FROM CLIENT  
    WHERE CLIENT.C_ID = p_C_ID;  
END;  
/
```

Blacklisted clients table after:

C_ID	F_NAME	L_NAME	DOB	CITY	GENDER	REASON	ADDED_BY	ADDED_DATE
1	5 Carol	Davis	03-08-19	Phoenix	F	Excessive cancellations	Admin	24-11-01